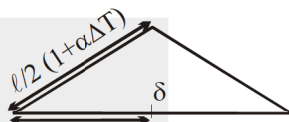


HINTS AND SOLUTIONS

Q.1 A

Sol. $\delta^2 = (\ell/2(1 + \alpha\Delta T))^2 - (\ell/2)^2$



Q.2 A

Q.3 B

Q.4 B

Q.5 C

Sol. $dQ = msdT = \frac{100}{1000} \int_{20}^{40} (100T + 500) dT$

$$= 10 \left(\frac{T^2}{2} + 5T \right)_{20}^{40}$$

$$= 10 \left[\left(\frac{40^2 - 20^2}{2} \right) + (40 - 20)5 \right]$$

$$= 10 \left[\frac{60 \times 20}{2} + 100 \right]$$

$$= 10 \times 700 = 7000 \text{ J}$$

Q.6 A

Sol. Applying newton's law of cooling

$$\frac{52 - 36}{10 \text{ min}} = \frac{-b}{mc} \left[\frac{52 + 36}{2} - 20 \right]$$

$$\alpha \frac{36 - \theta}{10 \text{ min}} = \frac{-b}{mc} \left[\frac{36 + \theta}{2} - 20 \right]$$

solving we get $\theta = 28^\circ\text{C}$

Q.7 B

Sol. $\frac{K_0 A (2T_0 - T)}{\ell} = \frac{d\theta}{dt}$

$$\frac{d\theta}{dt} = \frac{KA dt}{dx} \int \frac{dx}{k_0 \left(1 + \frac{x}{\ell} \right)} \frac{d\theta}{dt} = A \int_{T_0}^T dT$$

$$\frac{\ell}{k_0} \ln \left(1 + \frac{x}{\ell} \right) \Big|_0^{2\ell} \frac{d\theta}{dt} = A(T - T_0)$$

$$\frac{d\theta}{dt} = \frac{K_0 A}{\ell} (T - T_0) = \frac{K_0 A}{\ell} (2T_0 - T)$$

$$T - T_0 = 2T_0 \ln 3 - T \ln 3$$

$$T = \frac{(2\ell \ln 3 + 1)T_0}{1 + \ell \ln 3}$$

Q.8 C

Sol._{RD} Greater elasticity means less bending. More vertical thickness also implies less bending

Q.9 A

Q.10 A

Sol._{santosh} ∴ Intensity is inversely proportional to the square of the distance from the source.

$$\text{i.e. } I \propto \frac{1}{r^2}$$

$$\therefore \frac{I_{\text{earth}}}{I_{\text{mercury}}} = \frac{r_{\text{sm}}^2}{r_{\text{SE}}^2} = \frac{(0.4 r_{\text{SE}})^2}{(r_{\text{sp}})^2}$$

$$I_{\text{mercury}} = \frac{I_{\text{earth}}}{0.16} = \frac{2}{0.16} \frac{\text{cal}}{\text{min} - \text{cm}^2}$$

$$= \frac{200}{16} = 12.5 \frac{\text{cal}}{\text{min} - \text{cm}^2}]$$

Q.11 C

$$\text{Sol. } \frac{V_S}{V_T} = \frac{\rho_b}{\rho_L} = 0.8$$

$$\Delta T = 50^\circ\text{C}$$

$$\frac{V'_S}{V'_T} = \frac{\rho'_b}{\rho'_L} = \frac{\rho_b}{\rho_L} \times \frac{1 + \gamma_L \Delta T}{1 + \gamma_b \Delta T} = 0.8 \times \left(\frac{1 + 18 \times 10^{-5} \times 50}{1 + 5 \times 10^{-5} \times 50} \right) = 0.8 \times \left(\frac{1 + 9 \times 10^{-3}}{1 + 3 \times 10^{-3}} \right) = 0.8 \times (1 + 6 \times 10^{-3})$$

$$\frac{V'_T - V'_S}{V'_T} = 1 - 0.8 - 0.8 \times 6 \times 10^{-3} = 0.2 - 4.8 \times 10^{-3} = 10^{-2} [20 - 0.48] = 19.52\%]$$

Q.12 B

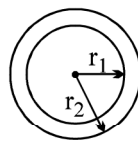
Sol._{OPC} Heat required by ice to convert in 100°C water will be $Q_1 = 1 \times 80 + 1 \times 100 = 180$ cal.
Steam will supply 180 cal heat and convert in water of 100°C . Temperature of mixture will be 100°C .

Q.13 D

$$\text{Sol. } \int dR = \int \frac{dr}{4\pi r^2 K} = \frac{1}{4\pi K} \left[\frac{r_2 - r_1}{r_2 r_1} \right] \quad [R = \text{thermal resistance}]$$



$$\dot{Q} = \frac{4\pi K \Delta \theta}{\left(\frac{r_2 - r_1}{r_1 r_2} \right)} = \frac{4\pi K \Delta \theta}{\left(\frac{t}{r^2} \right)}$$



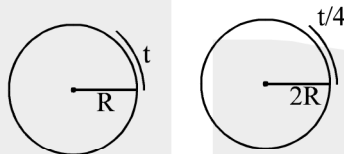
$$\frac{mL}{\text{time}} = \frac{4\pi K \Delta \theta}{\left(\frac{t}{r^2} \right)} \quad \left[m = \rho \times \frac{4}{3} \pi r^3 \right]$$

$$\frac{\rho L}{\text{time}} = \left(\frac{K}{t r} \right) \times \text{constant}$$

$$\text{time} \propto \frac{t r}{K}$$

$$\frac{25}{16} = \frac{\frac{t}{4} 2r K_s}{t r K_L} = \frac{1 K_s}{2 K_L}$$

$$\frac{K_L}{K_s} = \frac{8}{25}$$



Q.14 B

Q.15 C

Sol. $10 = \sigma e A [(313)^4 - (293)^4] \dots (1)$

$E = \sigma e A [(323)^4 - (293)^4] \dots (2)$

$E = \sigma e A [3514489440]$

$$E = \left[\frac{10}{[(313)^4 - (293)^4]} \right] \times [3514489440]$$

$$E = \left(\frac{10}{2227874160} \times 3514489440 \right)$$

Total heat loss to the surrounding will be = $\frac{10 \times 3514489440}{2227874160} \simeq 4732.52 \text{ J Ans.}$

Q.16 B

Q.17 C

Sol. $\frac{dQ}{dt} = \frac{kA}{L} (T_2 - T_1)$

$$65 = \frac{k \times 1.3}{18 \times 10^{-3}} (T - T_1) = \frac{1 \times 1.3}{4 \times 10^{-3}} (13 - T)$$

$$\Delta T_{\text{glass}} = 13 - T = 0.2$$

$$T = 12.8$$

$$\frac{18 \times 10^{-3} \times 50}{1.3} = k (12.8 - 0.2)$$

$$k = \frac{18 \times 10^{-3} \times 500}{12.6} = \frac{1}{14}$$

Q.18 BD

Q.19 C

Q.20 BC

Sol. $\frac{dQ}{dt} = eS \times A = ms \frac{dT}{dt}$

$\rightarrow \frac{dT}{dt}$ is more in spirit

In steady state

$$eSA = e \times \sigma (T_4 - T_0^4)$$

$\Rightarrow$ T is same in both.

Q.21 ABC

Q.22 AC

Sol. $I = \frac{MR^2}{2}$ & M is constant $\Rightarrow I \propto R^2$

$$\therefore \frac{\Delta I}{I} = 2 \left(\frac{\Delta R}{R} \right) = 2 \propto (\Delta T) \text{ on heating}$$

also no external torque present

$\therefore L = I\omega = \text{constant}$

$$\therefore \omega \propto \frac{1}{I}$$

$$\therefore \frac{\Delta \omega}{\omega} = - \frac{\Delta I}{I} = -2 \left(\frac{\Delta R}{R} \right) = -2 \propto (\Delta T)$$

Q.23 CD

Sol: $Q = \frac{KA(\Delta T)}{x} = \frac{KA(0 - (\theta))}{x}$; $\frac{d\theta}{dt} = \frac{KA\theta}{x}$

Q.24 ABD

Sol. (A) $15 \times \frac{1}{2} \times (0 - (-20)) = 150 \text{ cal}$

$$15 \times 80 = 1200 \text{ cal}$$

To melt we require 1350 cal

$$1350 + 15 \times 1 \times (T - 0) = 150 \times 1 \times (30 - T)$$

$$90 + T = 300 - 10T$$

$$T = \frac{210}{11} ^\circ\text{C}$$

(B) $5 \times 540 + 5 \times 1 \times (100 - T)$

$$= 150 \times 1 \times (T - 30)$$

$$3200 + 4500 = 155 T$$

$$T = \frac{1540}{31} ^\circ\text{C}$$

(C) $150(T - 30) = 80(70 - T)$

$$230T = 10100$$

